MFDB-Net: Multi-Attention Fusion Dual-Branch Network for Pavement Crack Detection

Zhong Qu, Tong Lu, Xue-Hui Yin, and Jian-Dong Wang

***Abstract*— Cracks are the most common disease in road maintenance work. They will accelerate the aging and damage of the pavement and bring potential risks to driving safety. Automatic detection of pavement cracks ensures infrastructure road safety. However, automatic detection of pavement cracks has always faced many challenges, such as irregular crack shapes and background noise interference. The existing methods ignore the correlation between cracked pixels, and there are some problems such as loss of detail information, lack of feature fusion and semantic gap. A Multi-Attention Fusion Dual-Branch Network(MFDB-Net) is proposed to accurately detect complete cracks. In this network, a dual-branch network is used as the backbone. A deeper branch extracts the semantic information of cracks, and a shallower branch retains more crack detail information. In addition, multiscale features are extracted by connecting multiple dilated convolutions in series with different convolution kernel sizes, which establishes global context connections. The attention mechanism is introduced to extract high-level network features and low-level network features respectively. Fusing features brings into play the characteristics of different network layers. Five publicly available crack datasets are used in our experiments, and the assessment is performed based on six evaluation metrics. We achieve *F1-score* of 0.879 and *MIoU* of 0.886 on the DeepCrack, *F1-score* of 0.737 and *MIoU* of 0.775 on the Crack500, *F1-score* of 0.571 and *MIoU* of 0.698 on the CrackTree200, and *F1-score* of 0.602 and *MIoU* of 0.712 on the EdmCrack600. We also carry out the generalization experiments on CFD. Sufficient experiments demonstrate our method present superior performance compared to other methods.**

***Index Terms*— Crack detection, dual-branch network, dilated convolution, multi-attention fusion.**

I. Introduction

CRACKS are one of the most common diseases in highway maintenance work, and they have various causes. Factors such as ground subsidence, harsh climatic conditions, and long-term traffic load can affect the pavement and cause cracks. Cracks will pose potential risks to driving safety. Therefore, rapid and accurate crack detection is particularly important for repairing crack.

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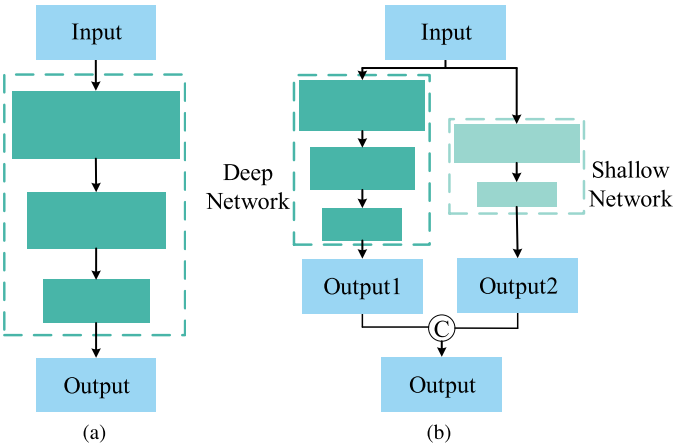

Fig. 1. Design of dual-branch network. © is Concatenate operation. To compare common networks to our dual-branch network. (a) Common crack detection network. (b) Our dual-branch network.

With the development of digital image technology, automatic crack detection methods have been proposed. Some methods have considered the difference between crack pixel values and background pixel values in pavement images, such as threshold, minimum path selection (MPS), and manually designed feature descriptors [1], [2], [3], [4], [5].

Although the above methods use the illumination, edge, shape and other characteristics of crack images, they are sensitive to background noise and ignore the correlation between crack pixels, which is prone to false detection, missed detection, and poor robustness. These methods are usually suitable for simple background scenes.

The rapid development of deep learning in recent years has provided more methods for detecting road cracks. Deep learning models learn rich and complex crack features by aggregating different convolutional features through pixel-by-pixel segmentation. Deep learning models have high accuracy and robustness and can effectively adapt to crack detection tasks in various scenarios. Although some results have been achieved, there is still potential for improvement in the refinement of deep semantic information and the retention of detail information, which deserves further exploration and research.

On this basis, we propose a novel dual-branch network as the backbone structure. As shown in Fig. 1, shallow networks with fewer network layers and smaller receptive fields retain rich detailed information, while deep networks with more network layers and larger receptive fields extract deeper semantic information. To enhance the network's ability

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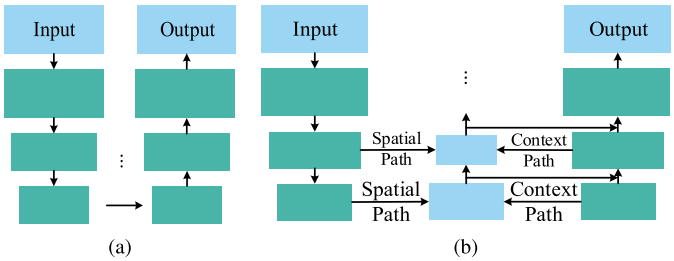

Fig. 2. Comparison of U-shaped network feature mapping methods. (a) Direct feature mapping method. (b) Our proposed feature mapping method.

to extract intricate spatial details and encompass a larger receptive field, we employ a dilated method within the decoder phase of the deep neural network. This approach effectively enlarges the receptive field, maintaining the original resolution of the features. As shown in Fig. 2, some common methods directly map low-level encoding network features to high-level decoding network features to achieve feature recovery, but ignore the semantic gap between encoding and decoding. We abandon the direct mapping method and use the spatial path to obtain rich spatial information in the low-level network, and use the semantic path to extract semantic information from the high-level network, fusing the two features as the side output and feature input of the next layer of the network. We fully leverage the characteristics of network layers at all levels to achieve improved integration of encoding and decoding functions. The main contributions of this paper are as follows,

(1) A novel dual-branch network is proposed, which uses asymmetric U-shaped structure as the backbone of the deep branch and multiscale strip convolution as the main layer of the shallow branch. The dual-branch networks are trained together and the features of the two branches are finally fused. The network can retain detailed information while extracting deep semantic information.
(2) We design series dilated residual modules that are used to extract multi-scale features and fuse local short-range feature information and complex long-range feature information. By continuously stacking multiple parallel combinations of ordinary convolutions and dilated convolutions with different dilation rates, the receptive field is continuously increased while keeping the size of the feature map unchanged.
(3) The multi-attention fusion module is inserted between the feature mappings of the U-shaped network to extract the spatial features of the low-level network and the semantic features of the high-level network, effectively utilizing the unique properties of each network layer and reducing the feature mapping gap between different network layers.

The following organization of this paper is as follows. Sec. II introduces previous research works. The details of our proposed network are described in Sec. III. Sec. IV demonstrates the effectiveness of our proposed method compared to other methods. Sec. V is the conclusion of this paper.

II. Related Work

In this section, we briefly overview a variety of techniques related to crack detection in the field of computer vision, aiming to improve the accuracy of crack detection.

A. Dual-Branch Network

Yu et al. [6], [7] proposed a bilateral segmentation network BiSeNet, which uses spatial path to retain spatial information, and context path to obtain larger receptive fields. Hong et al. [8] proposed a network DDRNet, which enhances feature extraction by extensively information between two branches through bilateral fusion. Sun et al. [9] proposed a lightweight network, which enhances the segmentation performance of the far end of the image while retaining detailed information. Zhang et al. [10] proposed a dual-stream network that combines local and global branches to enhance feature extraction. Xu and Xiong [11] proposed a three-branch network PIDNet, which is composed of detail path, context path and edge information path, by establishing the connection between convolutional network and proportional integral differential controller. Yadav et al. [12] proposed a dual-branch transformer structure to add self-attention information along the semantic and pixel channels. Xu et al. [13] proposed a learning model that fuses binary mask stream and high-resolution stream to obtain better. Gao et al. [14] adopted a dual attention module to fuse multimodal features. Yuan et al. [15] used two branches to process features of different modalities independently to make predictions for density maps and bounding boxes.

B. Dilated Strategy

Jiang and Liu [16] used a hybrid dilated convolutional block enhances the receptive field and uses fewer downsampling operations to reduce the loss of features. Ye et al. [17] replaced ordinary convolutions with dilated convolutions, which increased the receptive field in the shallow layers of the network. Zhou et al. [18] used cascaded multiscale dilated convolutions at the center of the network, and obtained multiscale information without additional parameters. Song et al. [19] proposed an extended convolutional attention module that captures multiscale contextual information from different regions of the image by locking multiple regions with receptive fields of different sizes. Wang et al. [20] incorporated dilated convolutions to integrate channel attention and spatial attention as mechanisms for enhancing features. Chen et al. [21] integrated features of different receptive fields through atrous convolution, solving the network's insensitivity to changes in target scale. Zhang et al. [22] effectively extracted multiscale features by inserting multiple different dilated convolutions. Gao et al. [23] proposed a sequence-to-image module to extract global information.

C. Attention Mechanism

Hu et al. [24] proposed the SE (Squeeze-And-Excitation) module to model the interdependence between convolutional feature channels. Qu et al. [25] used cascade and parallel dilated convolutions to expand the receptive field of feature points without reducing the resolution of feature maps. Yang et al. [26] proposed to extract attention information in three dimensions (space, channel and pixel) simultaneously, which is beneficial to extracting small cracks and suppressing the interference of background information. Hang et al. [27] used vertical and horizontal compressed attention modules to generate integrated pixel-level feature maps, and used channel attention upsampling modules to provide context information

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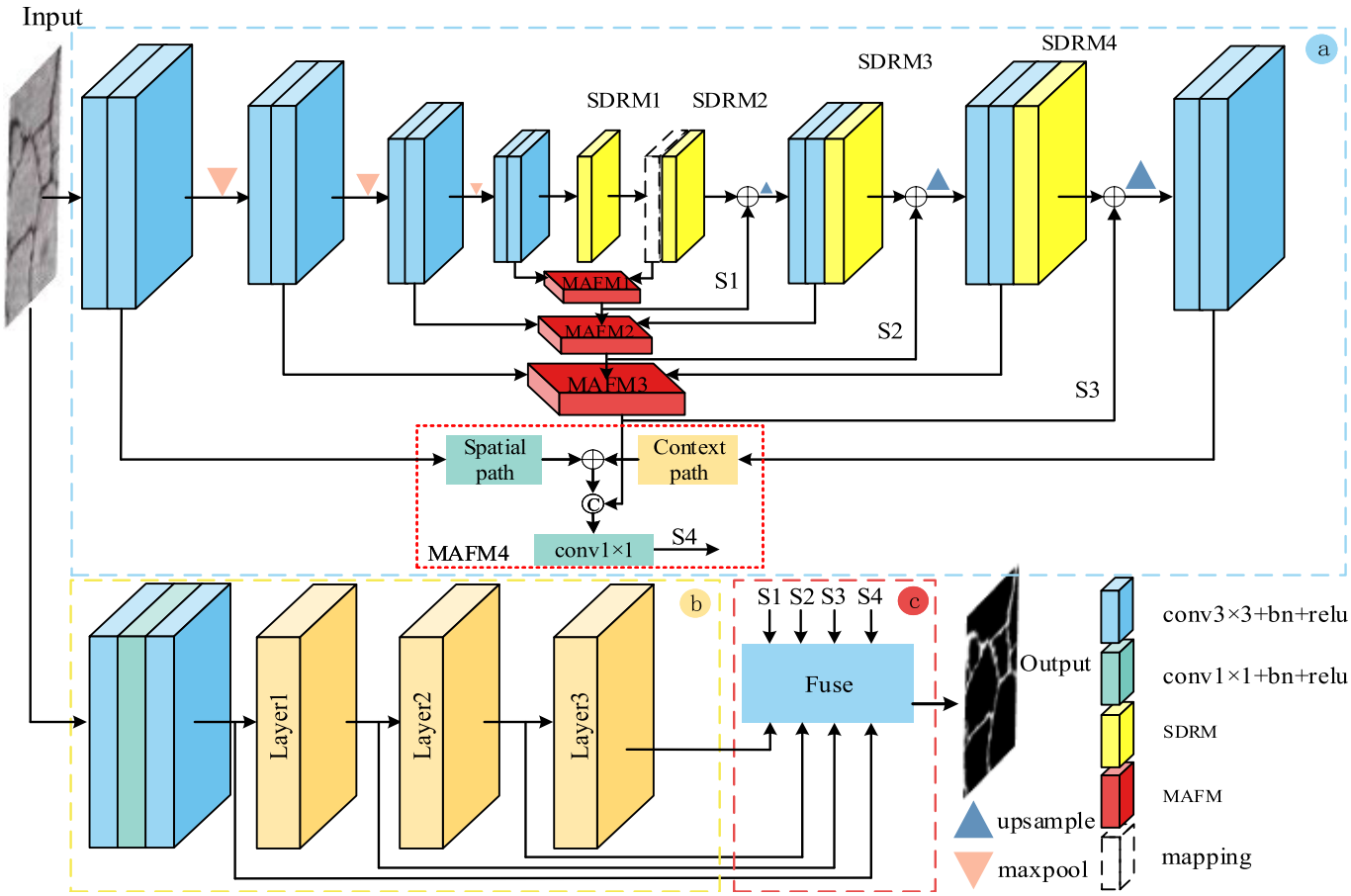

Fig. 3. Architecture of the proposed MFDB-Net. The network consists of a dual-branch network architecture (DB), four series dilated residual modules (SDRM), and four multi-attention fusion modules (MAFM). Part a: Deep network for extracting deep-level semantic feature. Part b: Shallow network for preserving spatial feature. Part c: Fusion of features at different levels. Layer is a multiscale strip convolution residual layer. Mapping is to directly map the result of the previous layer without processing. S_i is the side output of the deep network.

as guidance to help features recover spatial information. Zhou et al. [28] designed a multiscale attention spatial pyramid pooling module to extract features of different scales. Zhu et al. [29] used dual-layer routing to implement dynamic sparse attention, further achieving a heightened ability to allocate computations with better awareness of content.

III. The Proposed Methods

This section introduces our proposed network MFDB-Net. Following that, we introduce the dual-branch network, the series dilated residual module, and the multi-attention fusion module. Finally, we describe the loss function.

A. Network Architecture

Crack detection is a semantic segmentation task. The proposed network will perform pixel-wise binary classification, which classifies each pixel as a cracked pixel or a non-cracked pixel. Fig. 3 shows our proposed MFDB-Net network architecture in detail, which consists of three main parts: (1) Backbone network DB (dual-branch network) for crack feature extraction; (2) A series dilated residual module that can extract multiscale features. (3) Multi-attention fusion module that can fully integrate network features at all levels. Firstly, DB is a novel dual-branch network for crack detection. DB consists of a shallow network and a deep network. The shallow network retains rich crack detail information, and the deep network extracts more refined semantic information. Secondly, we add series dilated residual module and multi-attention fusion module to the deep network. The former extracts multi-scale crack features and establishes global context connections. The latter integrates network layer features at all levels and effectively narrows down the semantic gap that exists between low-level network features and high-level network features. Finally, the feature information extracted by the two branch networks is fused to obtain more complete crack characteristics. Sigmoid activation function is used to generate a probability map of the entire image for crack prediction.

B. Dual-Branch Network Architecture

DB mainly consists of shallow network and deep network. In order to make the shallow network retain more detailed information, we only design four network layers, including one ordinary layer composed of three convolutional layers and three multiscale strip convolution residual layers. Inspired by Inception-v3 [30], we decompose N×N convolution into N×1 convolution and 1×N convolution, and design a multiscale strip convolution residual layer. As shown in Fig. 4, the multiscale strip convolution residual layer is composed of three decomposed convolution layers with different convolution kernel sizes (3, 5, 7) in parallel, which are added and fused with the input of the skip connection, and finally passed 1×1 convolutional layer increases the number of channels.

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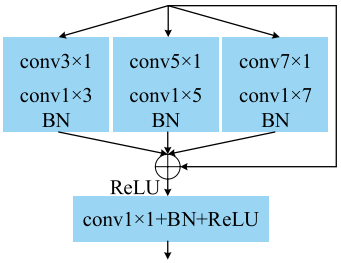

Fig. 4. MultiScale Strip Convolution Residual Layer. It is composed of multiple strip convolutions with different convolution kernel sizes in parallel. BN is Batch Normalization, $ReLU$ is nonlinear activation function.

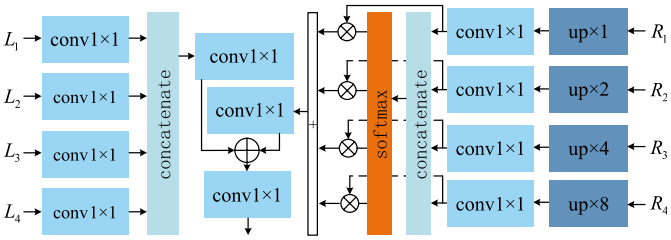

Fig. 5. Feature Fusion Module. The left part is the output feature processing of the shallow network, and the right part is the output feature processing of the deep network. Adaptively fuse the output features of shallow networks and deep networks. $L_i\,(i = 1, 2, 3, 4)$ is the side output of the shallow network, $R_i\,(i = 1, 2, 3, 4)$ is the side output of the network.

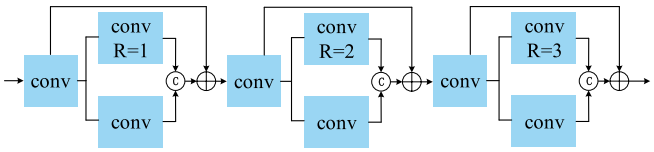

Fig. 6. Series Dilated Residual Module. conv is conv3 × 3, R is dilated rate, C is Concatenate operation, + is element-wise addition.

The 4 network layers of the shallow network provide four side outputs. In order to enable the deep network to extract deeper semantic information, we chose a U-shaped encoding and decoding network. It is divided into four feature extraction stages, each stage consists of corresponding encoding and decoding layers. The four stages of the deep network also provide 4 side outputs. As shown in Fig. 5, the features extracted by the shallow network and the deep network are fused. The output of the shallow network preserves the edge information of the crack. In order to retain information as much as possible, we use 1×1 convolution to reduce the number of side output channels, and then splice along the channels to obtain shallow features. Inspired by dynamic feature fusion [31], we first upsample the side output of the deep network and restore it to the original image size, then use 1×1 convolution, concatenate, and softmax to obtain the weight of each side output, and then assign the weight to the corresponding output to get weighted features. Four weighted features are added and fused, and after 1×1 convolution, deep features are obtained. Finally, the shallow features and deep features are added and fused, and passed through 1×1 convolution.

C. Series Dilated Residual Module

While increasing the feature receptive field, ordinary convolution layers and pooling layers reduce feature resolution and

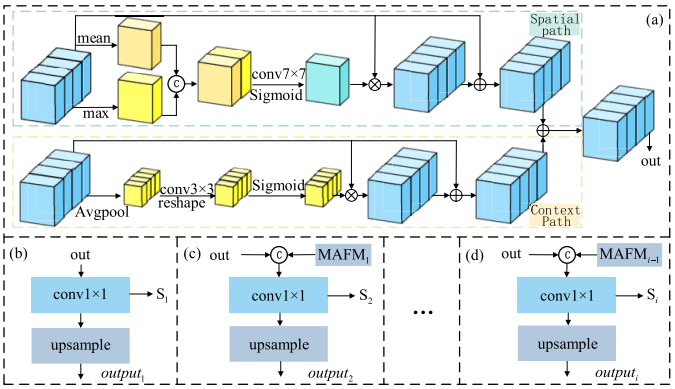

Fig. 7. Multi-Attention Fusion Module. out is the output of attention extraction block, S_i is the side output, output is the output of multi-attention fusion module, C is Concatenate operation. (a) Attention extraction block in multi-attention fusion module. (b) $MAFM_1$. (c) $MAFM_2$. (d) $MAFM_i$.

cannot meet the spatial information and larger receptive field required for crack detection. We design a feature extraction module based on multiscale dilated convolution. By concatenating multiple dilated convolution residual blocks with different dilated rates, it continuously increases the receptive field while keeping the size of the feature map unchanged. The obtained multiscale features capture global contextual information. The series dilated residual module is shown in Fig. 6. Firstly, we reduce the number of input feature channels through a 3×3 convolution layer. In order to extract multiscale features, parallel 3×3 convolutional layers and 3×3 dilated convolutional layers are used. Secondly, the multiscale features are spliced along the channels, and the number of channels is restored at the same time. The obtained multi-scale features are additively fused with the original input x of the residual jump. In order to better adapt to the size of the crack feature and solve the grid influence of dilated convolution, the dilated rate of the series dilated residual module is set to 1, 2, and 3. The above operations can be expressed as follows,

$$x_1 = \sigma(Bn(C_{3\times3}(x))), \tag{1}$$

$$y_1 = \sigma(Bn(C_{3\times3}(x_1))), \tag{2}$$

$$y_2 = \sigma(Bn(C_{3\times3,R}(x_1))), \tag{3}$$

$$y = Concat(y_1, y_2) + x, \tag{4}$$

where x is the input of SDRM, $C_{3\times3}$ is 3×3 convolution operation, Bn is Batch Normalization, σ is $ReLU$, R is dilated rate, $Concat$ is concatenate operation, x_1 is the output of the first layer 3×3 convolution, y_1 is the output of the second layer 3×3 convolution, y_2 is the output of the second layer 3×3 dilated convolution, y is the output of SDRM.

D. Multi-Attention Fusion Module

The attention mechanism allows the network to learn the areas that need attention in each image. The attention commonly used in crack detection does not fully leverage the characteristics of network layers, resulting in low utilization of crack features. The U-shaped encoding and decoding structure used in this study is a commonly used network architecture in semantic segmentation. Direct feature mapping of low-level networks and high-level networks can restore certain features, but there is still a large semantic gap. Inspired by CBAM [32] and ECA [33], as shown in Fig. 7, we design a feature fusion module based on a multi-attention mechanism. This module

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uses the spatial attention path to obtain the spatial information of the low-level networks, and uses the channel attention path to extract the semantic information of the high-level networks. By fusing two types of information, it improves the full utilization of the characteristics of different network layers and reduces the loss and redundancy of crack feature information. The low-level networks of the deep networks retain rich spatial information. We perform global average pooling and global maximum pooling on the low-level network features, and then splice the processed features along the channel. Through the 7×7 convolution operation and the Sigmoid activation function obtain the spatial attention. The spatial attention and the original features are multiplied and fused to obtain a weighted spatial matrix. Finally, the spatial feature map and the original features are added and fused to obtain the spatially weighted features. The above operations can be expressed as follows,

$$x_1 = \sigma(C_{7\times7}(Concat(Mean(x), Max(x)))) \times x + x, \quad (5)$$

where x is the input of the multi-attention fusion module, x_1 is the output of the spatial attention path, $Mean$ and Max represent the calculation of feature average and maximum values along the channel, $Concat$ is concatenate operation, $C_{m\times n}$ denotes convolution operation with the kernel size $m\times n$, σ is Sigmoid.

The high-level network of the deep network has rich semantic information. Firstly, average pooling operation is performed on the features along the channel. The channel attention is obtained through 3×3 convolution operation, dimension reshaping and Sigmoid activation function, and the channel attention is combined with the original features. The channel matrix with weight is obtained by multiplication and fusion, and finally the channel feature map and the original feature are added and fused to obtain the channel weight feature.

$$x_2 = \sigma(Reshape(C_{3\times3}(Avg(x)))) \times x + x, \quad (6)$$

where x is the input of the multi-attention fusion module, x_2 is the output of the channel attention path, Avg represents average pooling for each channel of the feature, $C_{m\times n}$ denotes convolution operation with the kernel size m×n, $Reshape$ represents changing the characteristic shape, σ is Sigmoid.

x_1 and x_2 contain spatial information and semantic information of crack features respectively, and the features with spatial weight and features with channel weight are added and fused to obtain a richer feature out.

$$out = x_1 + x_2, \quad (7)$$

We design four multi-attention fusion modules $MAFM_i$ ($i = 1, 2, 3, 4$). Except for the input of the first module is out, the inputs of the remaining three modules are additionally added with the output of the previous module $output_{i-1}(i = 1, 2, 3, 4)$.

When $i = 1$,

$$S_i = C_{1\times1}(out), \quad (8)$$

$$output_i = Up(C_{1\times1}(out)), \quad (9)$$

When $i = 2, 3, 4$,

$$S_i = C_{1\times1}(Concat(out, output_{i-1})), \quad (10)$$

$$output_i = Up(C_{1\times1}(Concat(out, output_{i-1}))), \quad (11)$$

where $Concat$ is concatenate operation, $C_{m\times n}$ denotes convolution operation with the kernel size m×n, S_i is a side output of the deep network, Up represents upsampling operation, $output_i$ is the output of $MAFM_i$.

E. Loss Function

Crack detection is a common problem and usually involves dividing them into cracked pixels and non-cracked pixels. Our training set is $S = (X^n, Y^n), n = 1, \ldots, N$, where sample X^n represents the crack image, Y^n represents the label image, and all parameters of the network are denoted W. Therefore, the side output loss function of the network is defined as follows,

$$\begin{aligned} L_{side}(X, Y, W) &= \sum_{m=1}^{n} \delta_m \Delta(X, Y, W) \\ &= \sum_{m=1}^{n} \delta_m \Delta(Pre_m, Y, W), \end{aligned} \quad (12)$$

where m represents the m_{th} side output layer, δ_m represents the loss weight hyperparameter of the m_{th} output layer, $Pre = \{Pre_i, i = 1, \ldots, |X|\}$, $Pre_i \in \{0, 1\}$ represents the prediction result of the m_{th} output.

In crack detection, there are far more non-crack pixels than crack pixels. We use a weighted cross-entropy loss function, Δ is defined as follows,

$$\begin{aligned} \Delta = &-W_0 \sum_{i\in G_+} \log \mathrm{P}(P_i = 1|X, W) \\ &- W_1 \sum_{i\in G_-} \log \mathrm{P}(P_i = 0|X, W), \end{aligned} \quad (13)$$

where $W_0 = \frac{|G_-|}{|G_+|}$ is the weight of the crack pixel, $W_1 = 1$ is the weight of non-crack pixels, G_+ is the total number of cracked pixels of the input image, G_- is the total number of non-cracked pixels of the input image, P is the probability that a pixel is predicted to be cracked or non-cracked. $P_i = 1$ means this pixel is crack, $P_i = 0$ means this pixel is non-crack.

After each side output is fused, the fusion prediction is obtained. The fusion loss function is defined as follows,

$$\begin{aligned} L_{fuse} = &-\delta W_0 \sum_{i\in G_+} \log \mathrm{P}(P_i = 1|X, W) \\ &- \delta W_1 \sum_{i\in G_-} \log \mathrm{P}(P_i = 0|X, W), \end{aligned} \quad (14)$$

where the parameters are the same as formulas (12) and (13). The proposed network architecture consists of a deep network and a shallow network. In order to make the semantic information extracted by the deep network richer, we only calculate the side output loss function of the deep network. The overall loss function is defined as follows,

$$\begin{aligned} L = &L_{fuse}(X, Y, W) + L_{fuse1}(X, Y, W) \\ &+ L_{fuse2}(X, Y, W) + L_{side}(X, Y, W), \end{aligned} \quad (15)$$

where L_{fuse} represents the deep network and shallow network feature fusion loss function, L_{fuse1} represents the side output fusion loss function of the deep network, L_{fuse2} represents the side output fusion loss function of the shallow network, L_{side} represents the side output loss function of the deep network.

IV. Experiments and Results

Fig. 8 and Fig. 9 present the results of nine methods on datasets. It illustrates the exceptional capabilities of the proposed method in identifying small cracks accurately and reducing background interference. This section provides the experimental instructions, including an overview of the datasets, a description of the comparison method

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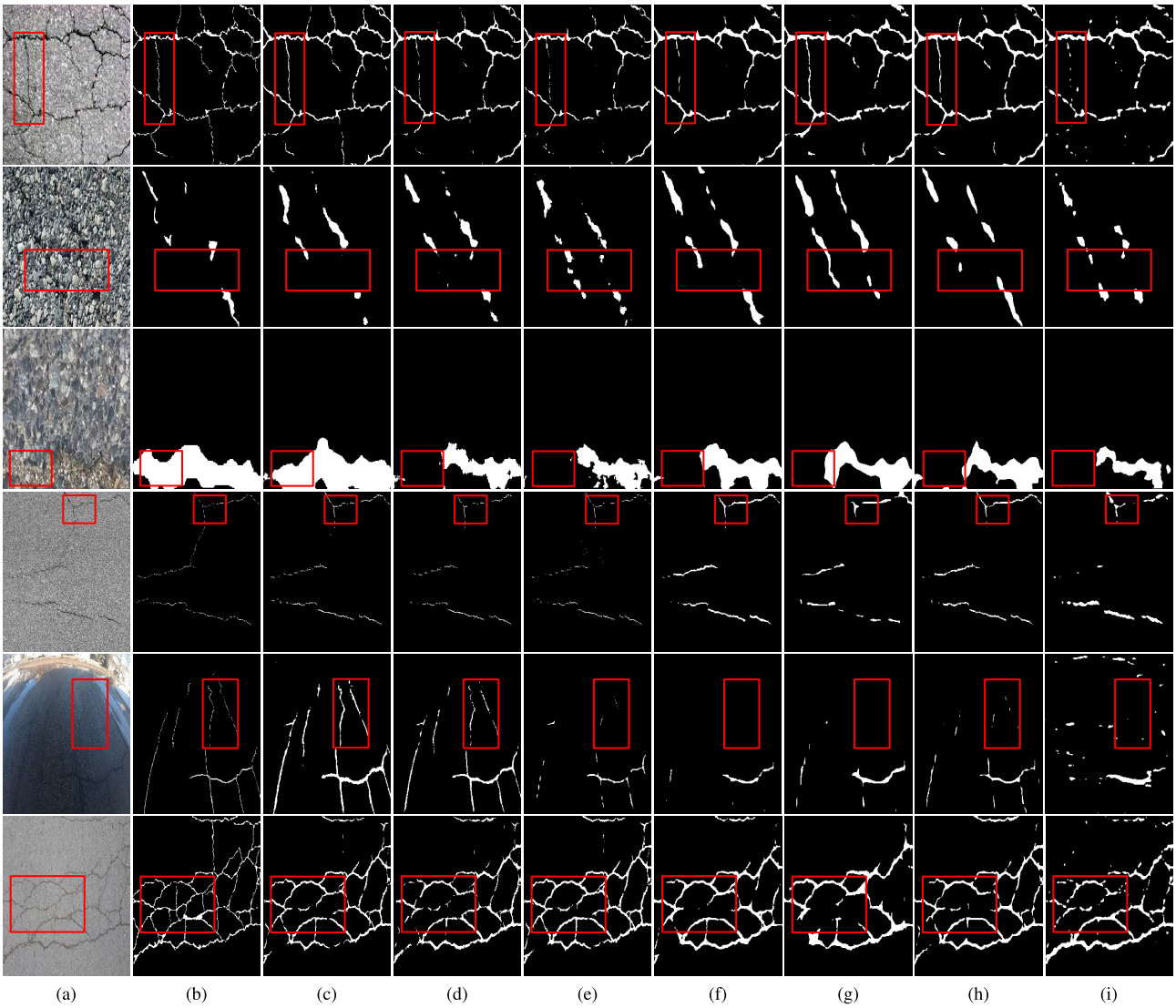

Fig. 8. The visualization of detection results. (a) Raw image; (b) Ground truth; (c) MFDB-Net(Ours); (d) DB(Ours); (e) U-Net; (f) DeepCrack; (g) FPHBN; (h) UHDN; (i) CrackW-Net. From top to bottom are DeepCrack, Crack500, Crack500, CrackTree200, EdmCrack600, CFD.

and evaluation criteria employed in the experiment. In the end, a detailed analysis and presentation of the experimental findings is provided.

A. Implementation Details

We use the PyTorch framework to implement the proposed network architecture. During the training process, the batch size is 1 (The batch size of Crack500 is 4), the training epoch is 200, the learning rate is 0.0001, the learning rate strategy decreases by a factor of 10 after every 40 epochs. We choose the SGD optimizer with a momentum of 0.9 and weight decay of 0.0002 to update network parameters as the optimization algorithm. Limited by GPU, we randomly adjust the size of the input image to a smaller dimension of 256×256 in order to shorten the training duration. All our experiments are performed on an NVIDIA A100-PCIE-40GB server.

B. Datasets

1) *DeepCrack* [36]: The dataset used in this paper consists of 537 images of concrete surface cracks, each image has a resolution of 384 × 544 pixels. The dataset includes different types of cracks such as bare cracks, coarse cracks and dirty cracks. We use 300 images to train the weight of the network and 237 images to test the superiority of the network.
2) *Crack500* [38]: The dataset consists of 500 photographs of complex pavement cracks that were obtained using a mobile device and annotated at the pixel level by Temple University. Each image has a resolution of 2000×1500, but due to limited processing power, the images were randomly cropped to 600×360 resulting in 3368 crack images. We use 1896 images to train the weight of the network, 1124 images to test the superiority of the network, and 348 images to validate.
3) *CFD* [44]: The dataset utilized in this paper consists of 118 images of road cracks on asphalt surfaces, each with a resolution of 480 × 320 pixels and manually labeled crack contours. The images in the dataset are unevenly illuminated and contain several sources of noise, including shadows, occlusions, and leaks, which can pose challenges for crack detection. We use this dataset for

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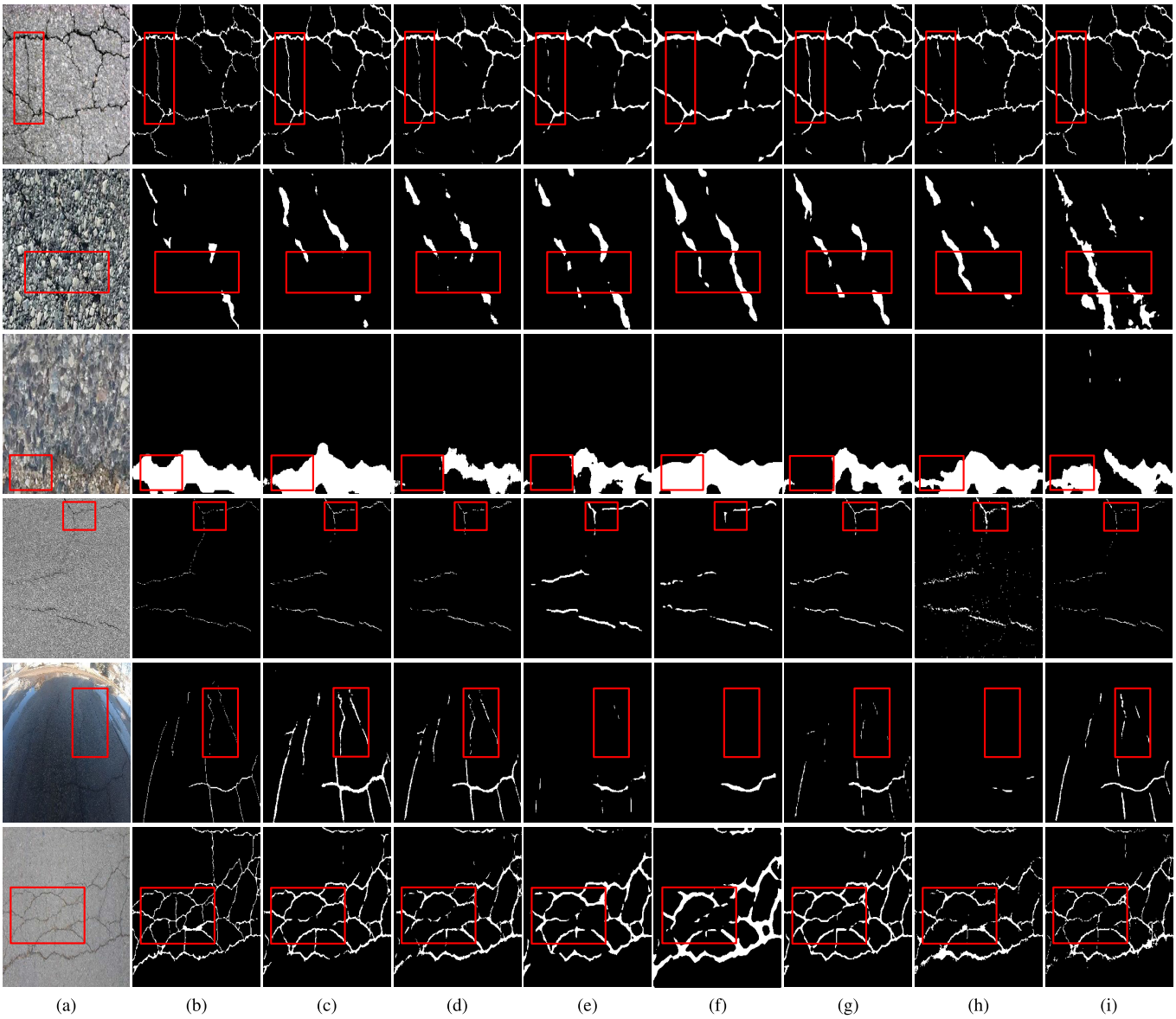

Fig. 9. The visualization of detection results. (a) Raw image; (b) Ground truth; (c) MFDB-Net(Ours); (d) DB(Ours); (e) MSCF; (f) CrackFormer; (g) PAF-Net; (h) DSC-Net; (i) CMUNeXt. From top to bottom are DeepCrack, Crack500, Crack500, CrackTree200, EdmCrack600, CFD.

generalization experiments to test the robustness of the network.

4) *CrackTree200* [45]: Zou et al. proposed a crack dataset named CrackTree200. This dataset consists of 206 road pavement crack images with a resolution of 800×600 pixels. A considerable portion of the visuals included in the dataset showcase minuscule cracks, reduced contrast, shadows, noise, and various other types of distortion. We use 166 images to train the weight of the network and 40 images to test.
5) *EdmCrack600* [46]: Mei et al. proposed a dataset called EdmCrack600, consisting of 600 fully annotated 1920×1080 pixel images. The data is designed to cover factors such as different weather conditions, different lighting conditions, the presence of shadows from other objects, and the difference in texture between different road surfaces. We use 420 images to train the weight of the network, 120 images to test the superiority of the network, and 60 images to validate.
6) *DCCE* [47]: Wang et al. selected 210 images from DeepCrack, Crack500, CFD, and EdmCrack600. DCCE contains transverse, longitudinal, grid-like, irregular crack shapes, shadows and occlusions. We use 140 images for training and 70 images for testing.

C. Comparison Methods

As shown in Table I, we conduct comparative experiments using some classic and recent crack detection networks (including some excellent semantic segmentation networks), and then we provide a clear introduction to these methods.

1) *U-Net* [34]: The design of U-Net neural network, characterized by its U-shape, integrates skip connections between the encoder and decoder structures. These connections enable the fusion of high and low-level features by linking downsampled feature maps from the encoder with upsampled feature maps of the same dimensions in the decoder.

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TABLE I
COMPARISON METHODS

Method	Journal	Algorithm	Year	Method	Journal	Algorithm	Year
U-Net [34]	Int. Conf. Med. Image Comput. Comput. -Assist. Intervent.	Convolutional networks for biomedical image segmentation	2015	MSCF [35]	*IEEE Trans. Intell. Transp. Syst.*	A lightweight network with multi-scale context fusion for real-time semantic segmentation	2022
DeepCrack [36]	*Neurocomputing*	A deep hierarchical feature learning architecture for crack segmentation	2019	CrackFormer [37]	*IEEE Trans. Intell. Transp. Syst.*	CrackFormer Network for Pavement Crack Segmentation	2023
FPHBN [38]	*IEEE Trans. Intell. Transp. Syst.*	Feature Pyramid and Hierarchical Boosting Network for Pavement Crack Detection	2020	PAF-Net [39]	*IEEE Trans. Intell. Transp. Syst.*	A Progressive and Adaptive Fusion Network for Pavement Crack Segmentation	2023
UHDN [40]	*Materials*	Automatic crack detection on road pavements using encoder-decoder architecture	2020	DSC-Net [41]	*IEEE Int. Conf. Comput. Vis.*	Dynamic Snake Convolution based on Topological Geometric Constraints for Tubular Structure Segmentation	2023
CrackW-Net [42]	*IEEE Trans. Intell. Transp. Syst.*	A novel pavement crack image segmentation convolutional neural network	2022	CMUNeXt [43]	*IEEE Int. Symp. Biomed. Imag.*	An Efficient Medical Image Segmentation Network Based on Large Kernel and Skip Fusion	2024

2) *DeepCrack* [36]: Liu et al. proposed a novel method for crack detection. To refine the prediction results, the method employs both bootstrap filtering and conditional random field (CRF) techniques.
3) *FPHBN* [38]: FPHBN utilizes a feature pyramid module and a hierarchical promotion module that is built upon the HED network, ultimately enhancing the network capacity for learning.
4) *UHDN* [40]: UHDN incorporates U-Net as its foundation, benefiting from both hierarchical feature learning and numerous expansion modules integrated into the network for effective identification of cracks.
5) *CrackW-Net* [42]: CrackW-Net is a pixel-level semantic segmentation network that expands the global receptive field through a skip-level round-trip sampling block structure, improving the ability to distinguish background, interference information, and segment objects.
6) *MSCF* [35]: An asymmetric encoding and decoding structure is proposed to capture multi-scale contextual information through efficient attention module branches at different stages of the network.
7) *CrackFormer* [37]: CrackFormer embeds the Transformer encoder block in a SegNet-like encoder-decoder structure. This encoder block has efficient relative position embedding and long-range interactions to extract effective contextual information from feature channels.
8) *PAF-Net* [39]: PAF-Net introduces progressive context fusion (PCF) blocks to capture context information from adjacent scales to mitigate information loss caused by feature downsampling. A dual attention (DA) block is proposed, which utilizes global and local context information to reduce the semantic gap problem.
9) *DSC-Net* [41]: DSC-Net proposes a multi-view feature fusion strategy and dynamic Snake convolution, which increases the attention to multi-view features in the feature fusion process to ensure that important information is retained under different global forms, and can adaptively focus on zigzag features.
10) *CMUNeXt* [43]: CMUNeXt utilizes a large kernel and reverse bottleneck design to thoroughly mix remote spatial and location information, effectively extracting global context information.
11) *DB*: DB is proposed dual-branch network without SDRM and MAFM. It consists of an asymmetric U-shaped deep network and a shallow network composed of multiScale strip convolution residual layers.

D. Evaluation Metrics

In this paper, we introduce $precision(P)$, $recall(R)$ and $F1{-}score(F_1)$ to evaluate these models. P, R, F_1 are calculated as follows,

$$P = \frac{TP}{TP + FP}, \tag{16}$$

$$R = \frac{TP}{TP + FN}, \tag{17}$$

$$F_1 = 2 \times \frac{P \times R}{P + R}, \tag{18}$$

where TP means true positive, FP means false positive, FN means false negative.

The crack detection is a binary classification. To further evaluate the proposed method, we introduce other evaluation metrics, such as Pixel Accuracy(PA), Class Average Accuracy(C) and Mean Intersection Over Union($MIoU$). They are calculated as follows,

$$PA = \frac{\sum_{i=0}^{n} t_{ii}}{\sum_{i=0}^{n} T_i}, \tag{19}$$

$$C = \frac{1}{n} \times \sum_{i=0}^{n} \frac{t_{ii}}{T_i}, \tag{20}$$

$$MIoU = \frac{1}{n} \times \sum_{i=0}^{n} \frac{t_{ii}}{T_i + \sum_{j=0}^{n} t_{ji} - t_{ii}}, \tag{21}$$

where n is the number of classes, t_{ij} represents the number of pixels of class i predicted as class j, T_i is the number of pixels of class i.

E. Experimental Results

1) Results on Deepcrack: As shown in the first row of Fig. 10, DeepCrack consists of cracks in the concrete pavement, which have distinct visual features. Compared with other crack datasets, DeepCrack has clear vision, wider crack width and less background interference. As shown in Fig. 11(a), MFDB-Net outperforms the highest *P-R* curve compared

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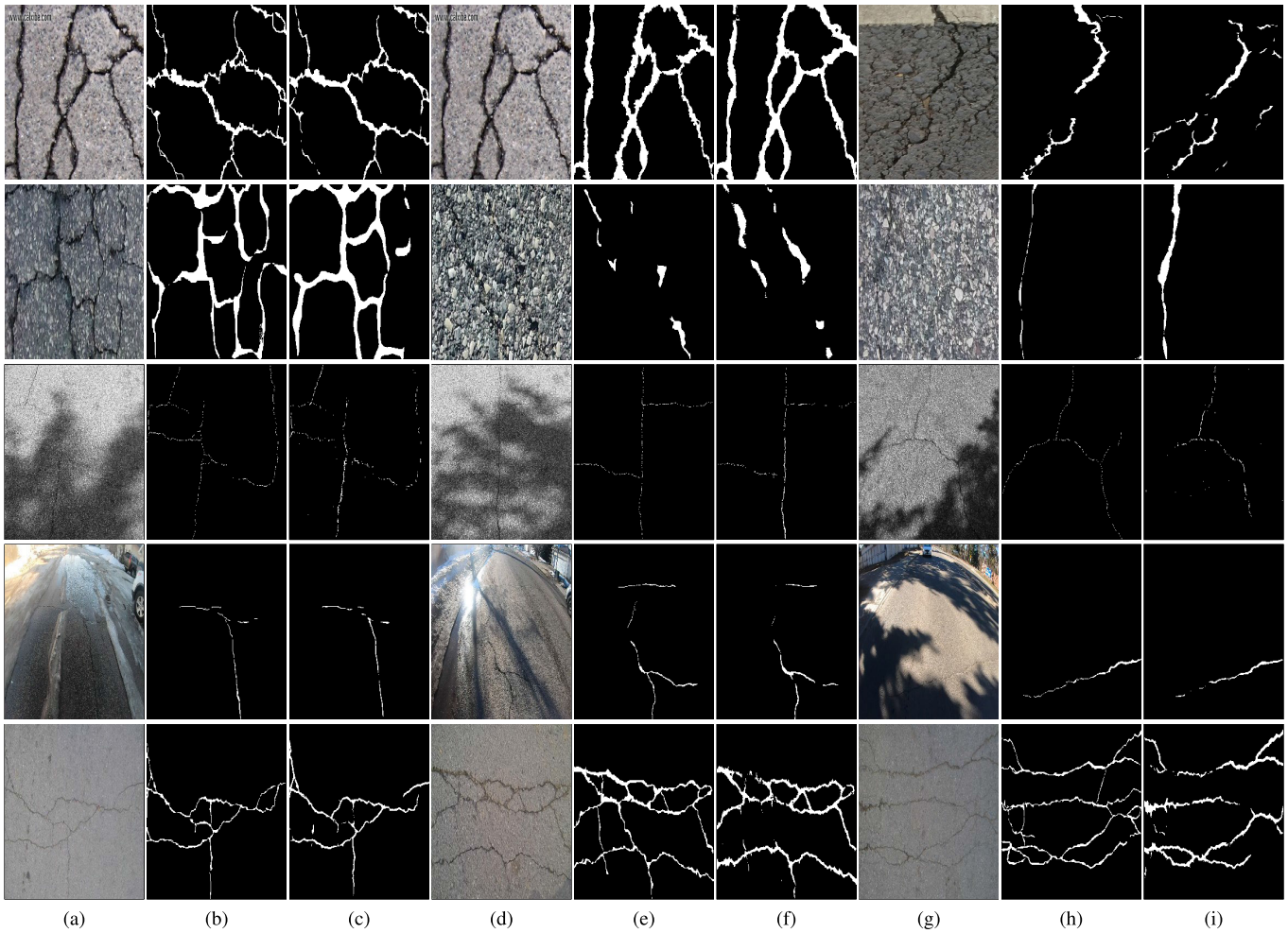

Fig. 10. The visualization of MFDB detection results on different datasets. (a)(d)(g) Raw image; (b)(e)(h) Ground truth; (c)(f)(i) MFDB-Net(Ours). From top to bottom are DeepCrack, Crack500, CrackTree200, EdmCrack600, CFD.

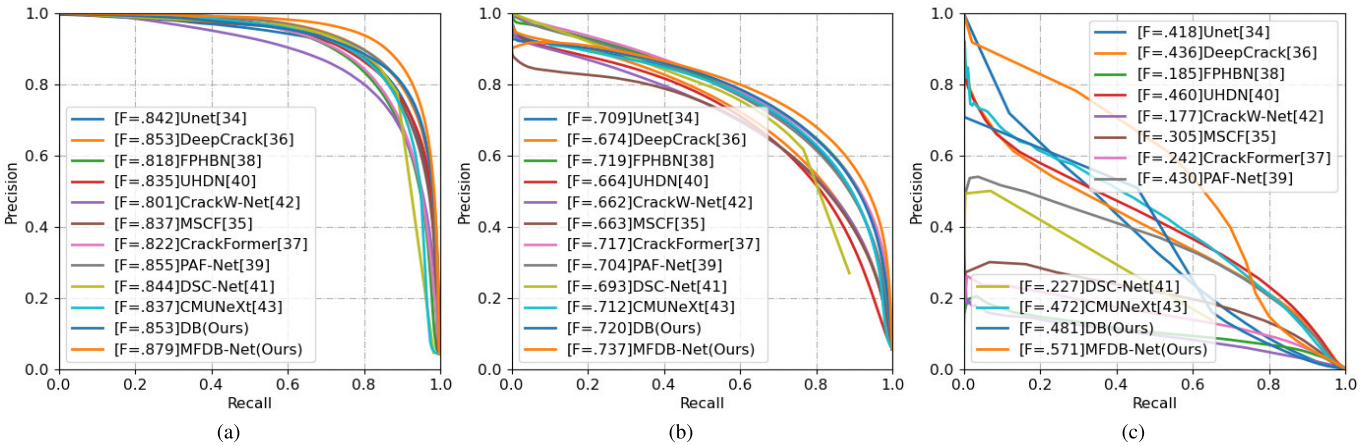

Fig. 11. The *P-R* curves on the three datasets. (a) DeepCrack. (b) Crack500. (c) CrackTree200.

to all other comparison methods, indicating its superiority. From Table II, MFDB-Net outperforms in all evaluation indicators, including P, R, F_1, PA, C and $MIoU$, each reaching the value of 0.889, 0.869, 0.879, 0.989, 0.932 and 0.886. Compared with U-Net, DeepCrack, MSCF, PAF-Net and DSC-Net, the F_1 performance improvements obtained by our method are 3.7%, 2.6%, 4.2%, 2.4% and 3.5% respectively. Compared with DB, the F_1 improves from 0.853 to 0.879 and the $MIoU$ improves from 0.864 to 0.886.

2) Results on Crack500: As shown in the second row of Fig. 10, Crack500 consists of cracks in the asphalt pavement. Compared with DeepCrack, the vision is blurred and the crack is less distinguished from the background. Fig. 11(b) depicts the PR curves of the compared methods

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TABLE II
COMPARISON RESULTS ON DeepCrack

Methods	Metrics					
	P	R	F_1	PA	C	$MIoU$
U-Net [34]	0.839	0.844	0.842	0.986	0.918	0.856
DeepCrack [36]	0.852	0.854	0.853	0.987	0.923	0.865
FPHBN [38]	0.833	0.802	0.818	0.984	0.897	0.837
UHDN [40]	0.831	0.839	0.835	0.985	0.915	0.851
CrackW-Net [42]	0.790	0.812	0.801	0.982	0.901	0.825
MSCF [35]	0.854	0.820	0.837	0.986	0.907	0.852
CrackFormer [37]	0.832	0.811	0.822	0.984	0.902	0.840
PAF-Net [39]	0.849	0.861	0.855	0.987	0.927	0.867
DSC-Net [41]	0.864	0.825	0.844	0.987	0.910	0.858
CMUNeXt [43]	0.846	0.828	0.837	0.986	0.910	0.852
DB(Ours)	0.827	**0.878**	0.853	0.986	**0.935**	0.864
MFDB-Net(Ours)	**0.889**	0.869	**0.879**	**0.989**	0.932	**0.886**

TABLE III
COMPARISON RESULTS ON Crack500

Methods	Metrics					
	P	R	F_1	PA	C	$MIoU$
U-Net [34]	0.679	0.742	0.709	0.966	0.856	0.757
DeepCrack [36]	0.651	0.697	0.674	0.962	0.837	0.734
FPHBN [38]	0.684	0.757	0.719	0.966	0.868	0.763
UHDN [40]	0.639	0.689	0.664	0.961	0.827	0.728
CrackW-Net [42]	0.628	0.700	0.662	0.960	0.838	0.726
MSCF [35]	0.624	0.707	0.663	0.960	0.834	0.727
CrackFormer [37]	0.682	0.755	0.717	0.966	0.867	0.762
PAF-Net [39]	0.674	0.737	0.704	0.966	0.850	0.754
DSC-Net [41]	0.673	0.714	0.693	0.965	0.840	0.747
CMUNeXt [43]	0.679	0.748	0.712	0.967	0.857	0.759
DB(Ours)	0.686	0.757	0.720	0.967	0.862	0.764
MFDB-Net(Ours)	**0.697**	**0.780**	**0.737**	**0.968**	**0.880**	**0.775**

on Crack500. The figure shows that the curve of the proposed method is the highest, indicating its superiority. As shown in Table III shows, our method obtains the best F_1 of 0.737 in Crack500. Compared with U-Net, FPHBN, CrackFormer, PAF-Net and CMUNeXt, the F_1 performance improvements obtained by our method are 2.8%, 1.8%, 2.0%, 3.3% and 2.5% respectively. Compared with DB, the F_1 improves from 0.720 to 0.737 and the $MIoU$ improves from 0.764 to 0.775.

3) Results on CrackTree200: As shown in the third row of Fig. 10, CrackTree200 is a concrete pavement crack with shadows, occlusion, low contrast, noise and other issues, compared to other data sets, the crack is extremely thin and irregular. As shown in Fig. 11(c), the overall performance of all methods is poor, because most of the CrackTree200 images are tiny cracks that are difficult to detect, but our method is still better than the compared methods. From Table IV, compared with the optimal comparison method, the proposed method outperforms it with 9.9% in F_1. Compared with DB, the F_1 improves from 0.481 to 0.571 and the $MIoU$ improves from 0.656 to 0.698.

4) Results on EdmCrack600: As shown in the fourth row of Fig. 10, EdmCrack600 is a dataset of asphalt pavement cracks, including noise such as changes in weather conditions, shadows, occlusion, stains, texture differences, and low contrast due to overexposure. As shown in Fig. 12, the overall performance of all methods is poor, because most of the EdmCrack600 images are tiny cracks that are difficult to detect, but our method is still better than the compared methods. From Table V, compared with the optimal comparison method, the proposed method outperforms it with 4.3% in F_1. Compared with DB, the F_1 improves from 0.525 to 0.602 and the $MIoU$ improves from 0.674 to 0.712.

5) Cross-Validation Results for MFDB: We conduct 5-fold cross-validation experiments, where datasets were divided into 5 subsets. As shown in Table VI, the DeepCrack training set has 300 images, and the mean of F_1 is 0.898, and the variance is 0.000169. The Crack500 training set has 2244 images,

TABLE IV
COMPARISON RESULTS ON CrackTree200

Methods	Metrics					
	P	R	F_1	PA	C	$MIoU$
U-Net [34]	0.407	0.429	0.418	0.995	0.713	0.630
DeepCrack [36]	0.354	0.565	0.436	0.995	0.781	0.637
FPHBN [38]	0.130	0.323	0.185	0.990	0.658	0.546
UHDN [40]	0.396	0.546	0.460	0.996	0.772	0.647
CrackW-Net [42]	0.121	0.321	0.177	0.991	0.627	0.543
MSCF [35]	0.219	0.499	0.305	0.993	0.747	0.586
CrackFormer [37]	0.172	0.408	0.242	0.991	0.700	0.564
PAF-Net [39]	0.369	0.515	0.430	0.995	0.756	0.635
DSC-Net [41]	0.138	**0.651**	0.227	0.985	**0.819**	0.556
CMUNeXt [43]	0.421	0.538	0.472	0.996	0.754	0.653
DB(Ours)	0.509	0.455	0.481	0.997	0.726	0.656
MFDB-Net(Ours)	**0.613**	0.532	**0.571**	**0.997**	0.766	**0.698**

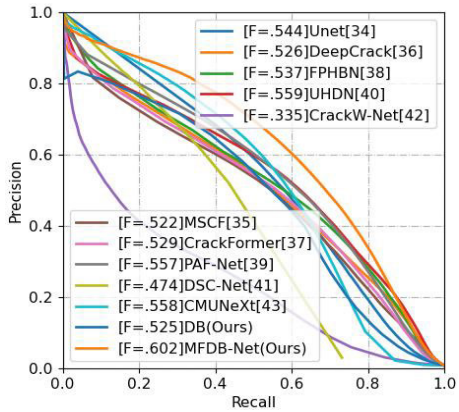

Fig. 12. The *P-R* curves on the EdmCrack600 dataset.

TABLE V
COMPARISON RESULTS ON EdmCrack600

Methods	Metrics					
	P	R	F_1	PA	C	$MIoU$
U-Net [34]	0.562	0.527	0.544	0.992	0.762	0.683
DeepCrack [36]	0.512	0.540	0.526	0.992	0.768	0.674
FPHBN [38]	0.483	**0.604**	0.537	0.992	0.782	0.679
UHDN [40]	0.534	0.587	0.559	0.992	0.791	0.690
CrackW-Net [42]	0.295	0.388	0.335	0.987	0.690	0.594
MSCF [35]	0.502	0.545	0.522	0.991	0.770	0.673
CrackFormer [37]	0.502	0.559	0.529	0.991	0.777	0.676
PAF-Net [39]	0.532	0.585	0.557	0.992	0.790	0.689
DSC-Net [41]	0.522	0.434	0.474	0.992	0.715	0.651
CMUNeXt [43]	0.592	0.527	0.558	0.993	0.762	0.690
DB(Ours)	0.539	0.510	0.525	0.992	0.753	0.674
MFDB-Net(Ours)	**0.608**	0.597	**0.602**	**0.993**	**0.797**	**0.712**

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TABLE VI
CROSS-VALIDATION EXPERIMENTS OF MFDB ON DIFFERENT DATASETS

Dataset	Metrics	
	$F_1_Average$	$F_1_Variance$
DeepCrack [36]	0.898	0.000169
Crack500 [38]	0.764	0.000081
CrackTree200 [45]	0.540	0.000194
EdmCrack600 [46]	0.599	0.000449

TABLE VII
COMPARISON RESULTS OF GENERALIZATION ON CFD

Methods	DeepCrack			Crack500		
	P	R	F_1	P	R	F_1
U-Net [34]	0.506	0.538	0.522	0.482	0.598	0.534
DeepCrack [36]	0.576	0.663	0.616	0.477	0.621	0.540
FPHBN [38]	0.540	**0.683**	0.604	0.439	**0.661**	0.527
UHDN [40]	0.564	0.657	0.607	0.490	0.645	0.557
CrackW-Net [42]	0.488	0.545	0.515	0.296	0.455	0.359
MSCF [35]	0.576	0.621	0.597	0.479	0.617	0.539
CrackFormer [37]	0.562	0.599	0.580	0.441	0.574	0.499
PAF-Net [39]	0.549	0.641	0.591	0.537	0.599	0.566
DSC-Net [41]	0.545	0.562	0.553	0.497	0.547	0.521
CMUNeXt [43]	0.493	0.587	0.536	0.423	0.584	0.490
DB(Ours)	0.555	0.624	0.588	0.506	0.618	0.557
MFDB-Net(Ours)	**0.594**	0.668	**0.629**	**0.557**	0.641	**0.596**

the mean of F_1 is 0.764, and the variance is 0.000081. The CrackTree200 training set has 166 images, the mean of F_1 is 0.540, and the variance is 0.000194. The EdmCrack600 training set has 480 images, the mean of F_1 is 0.599, and the variance is 0.000449. The overall effect of the model is consistent, stable, and has small fluctuations.

6) Results of Generalization on CFD: We conduct generalization experiments on CFD using networks trained by DeepCrack and Crack500 respectively. As shown in the fifth row of Fig. 10, CFD is the crack of cement pavement, and the crack features are small, including noise sources such as shadow, occlusion, oil spot and water stain. The crack is not distinguishable from the background, and the network detection is difficult. As shown in Table VII, The performance of all networks decreases as the difficulty of the dataset increases. Compared with the best comparison method, the proposed method achieves the best generalization performance, in which the network trained on DeepCrack achieves F_1 of 0.629 and the network trained on Crack500 achieves F_1 of 0.596.

7) The Visualization Results of Heat Maps of Different Methods: We present a visual comparison of the thermal maps of the proposed method MFDB with some other methods, as shown in Fig. 13(a), where the attention of MFDB is more focused on fracture characteristics. For example, U-Net(Fig. 13(b)), CrackW-Net(Fig. 13(e)) and other networks are difficult to distinguish between crack features and non-crack features, and will pay attention to some similar crack features. Heat maps demonstrate the robustness of our proposed method.

8) Results on Different Input Sizes: As shown in Table VIII, we conduct comparative experiments on the DeepCrack dataset with different input sizes. The proposed method MFDB-Net outperforms other methods under four different input sizes, and achieves the best performance at 256 × 256.

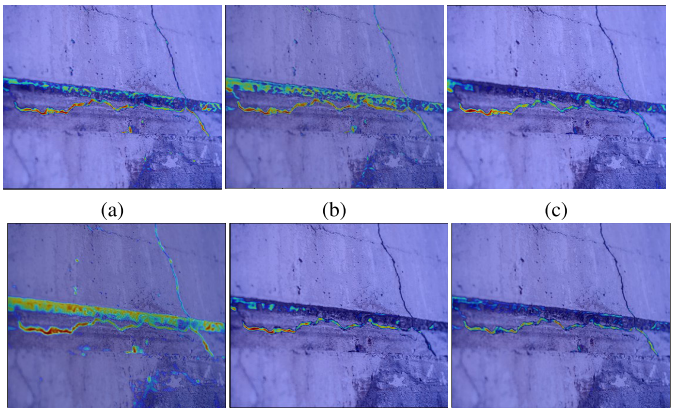

Fig. 13. The Visualisation results of heat maps of methods. (a) MFDB; (b) U-Net; (c) FPHBN; (d) CrackW-Net; (e) MSCF; (f) PAF-Net.

TABLE VIII
THE F_1 OF DIFFERENT INPUT SIZES ON DeepCrack

Methods	Input Sizes			
	128 × 128	256 × 256	384 × 384	512 × 512
U-Net [34]	0.806	0.842	0.848	0.840
DeepCrack [36]	0.819	0.853	0.843	0.811
FPHBN [38]	0.755	0.818	0.769	0.728
UHDN [40]	0.781	0.835	0.837	0.828
CrackW-Net [42]	0.776	0.801	0.815	0.814
MSCF [35]	0.782	0.837	0.841	0.842
CrackFormer [37]	0.620	0.822	0.821	0.820
PAF-Net [39]	0.787	0.855	0.835	0.834
DSC-Net [41]	0.825	0.844	0.833	0.820
CMUNeXt [43]	0.749	0.837	0.835	0.825
DB(Ours)	0.832	0.853	0.861	0.856
MFDB-Net(Ours)	**0.864**	**0.879**	**0.868**	**0.859**

TABLE IX
THE P, R, F_1 OF BRANCH NETWORKS

Threshold	Methods	Metrics		
		P	R	F_1
0.31	MFDB-Net_Shallow	0.756	0.916	0.828
	MFDB-Net_Deep	0.808	0.887	0.846
	MFDB-Net	0.889	0.869	0.879
0.37	MFDB-Net_Shallow	0.814	0.884	0.847
	MFDB-Net_Deep	0.870	0.842	0.856
	MFDB-Net	0.928	0.819	0.870
0.42	MFDB-Net_Shallow	0.855	0.850	0.852
	MFDB-Net_Deep	0.911	0.796	0.850
	MFDB-Net	0.951	0.768	0.850

The image size of the DeepCrack dataset is 384 × 544. 128 × 128 is too small for the original image, and 512 × 512 exceeds the original image size, so the detection results are poor. The comparison result of 384 × 384 is better than the previous two sizes, but it is still insufficient compared to 256 × 256.

9) Results in Branch Network: From Table IX, 0.31, 0.37, and 0.42 are the thresholds for MFDB-Net, MFDB-Net_Deep (the deep network branch of MFDB-Net), and MFDB-Net_Shallow (the shallow network branch of MFDB-Net) to obtain the best F_1 respectively. We find that the P of deep

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TABLE X
THE ABLATION EXPERIMENTS OF FEATURE FUSION MODULE ON DeepCrack

Methods	Metrics					
	P	R	F_1	PA	C	$MIoU$
FFM [6]	0.878	0.791	0.832	0.986	0.893	0.849
AFF [48]	0.879	0.862	0.870	0.988	0.928	0.879
BGA [49]	0.741	**0.882**	0.805	0.981	**0.934**	0.827
DEA [50]	0.855	0.860	0.857	0.987	0.926	0.869
EFC [51]	0.778	0.868	0.821	0.987	0.928	0.865
MFDB-Net(Ours)	**0.889**	0.869	**0.879**	**0.989**	0.932	**0.886**

TABLE XI
THE ABLATION EXPERIMENTS OF SDRM DILATED RATE ON DeepCrack

Dilated Rate	MAFM	Metrics		
		P	R	F_1
1,2,4	×	0.856	0.870	0.863
1,2,5	×	0.866	0.866	0.866
1,3,4	×	0.854	0.874	0.864
1,3,5	×	0.846	**0.880**	0.863
1,2,3	×	**0.867**	0.865	**0.866**
1,2,4	✓	0.884	0.861	0.863
1,2,5	✓	0.876	0.868	0.872
1,3,4	✓	0.877	0.864	0.870
1,3,5	✓	0.881	**0.870**	0.876
1,2,3	✓	**0.889**	0.869	**0.879**

networks is higher than that of shallow networks, and the R of shallow networks is higher than that of deep networks. This is because deep networks can extract more precise semantic information, so P is higher. Shallow networks can retain more detailed information, usually with higher R. These results are consistent with the original intention of designing the dual-branch network.

10) Ablation Experiments of Feature Fusion Module: As shown in Table X, compared with the existing feature fusion strategies, the proposed feature fusion module outperforms the other strategies. The feature fusion module can not only fully utilize the eight side outputs, but also make full use of the characteristics of DB, and adopt different methods for different feature outputs. In order to retain the edge information of the crack as much as possible, the shallow network output features are only operated by simple convolutional splicing to retain complete information. Deep networks are able to extract deeper semantic features. The output features of the deep network obtain feature weights through a series of operations, and then the weights are fused with the features to obtain features with richer semantics. The shallow network and the deep network have a total of eight side outputs, and it is crucial to effectively fuse eight side outputs.

11) Ablation Experiments of SDRM: SDRM consists of multiple series of dilated convolutions. We follow the basic principle [52] for the dilation ratio setting to avoid the generation of grid effects. As shown in Table XI, we conduct an ablation experiment on the dilation ratio. The dilated convolution is conducive to extracting multi-scale information and improving the semantic extraction ability of the network (P). However, too large a dilation ratio will make the sampling sparse and lose some local information, thus affecting the network's detail extraction ability (R). By balancing the P value and the R to obtain a higher F_1, we chose dilation ratio settings of 1, 2, and 3. We add SDRM in the four stages of the deep network branch. As shown in Table X, there are two groups of experiments in Table XII. The first group does not add MAFM, and the experimental results show that the F_1 is improved when only one SDRM is added. When we put all SDRMs in sequence, we get the highest P and F_1. Compared with the baseline DB, P increases by 4% and F_1 increases by 1.3%. The second group adds MAFM, and the results show that the designed modules are mutually reinforcing. In the first group of experiments, we do not add MAFM, which means that there is no feature mapping between the low-level networks and the high-level networks of the deep network. It is difficult to recover the lost features in the high-level network by upsampling alone. When MAFM is added, the high-level networks is compensated by the characteristics of the low-level networks, and SDRM can further play its role.

TABLE XII
THE ABLATION EXPERIMENTS OF SDRM ON DeepCrack

Methods	MAFM	Metrics		
		P	R	F_1
DB	×	0.827	0.878	0.853
DB+SDRM1	×	0.849	**0.879**	0.864
DB+SDRM2	×	0.861	0.866	0.864
DB+SDRM3	×	0.847	0.868	0.858
DB+SDRM4	×	0.840	0.873	0.856
DB+SDRM1_2	×	0.853	0.878	0.865
DB+SDRM1_2_3	×	0.859	0.875	0.864
DB+SDRM1_2_3_4	×	**0.867**	0.865	**0.866**
DB+SDRM1	✓	0.880	0.861	0.871
DB+SDRM2	✓	0.876	0.869	0.872
DB+SDRM3	✓	0.870	**0.871**	0.871
DB+SDRM4	✓	0.873	0.866	0.869
DB+SDRM1_2	✓	0.883	0.868	0.875
DB+SDRM1_2_3	✓	0.883	0.861	0.872
DB+SDRM1_2_3_4	✓	**0.889**	0.869	**0.879**

TABLE XIII
THE ABLATION EXPERIMENTS OF MAFM ON DeepCrack

Methods	SDRM	Metrics		
		P	R	F_1
DB	×	0.827	**0.878**	0.853
DB+MAFM1	×	0.842	0.874	0.858
DB+MAFM2	×	0.861	0.869	0.865
DB+MAFM3	×	0.857	0.865	0.861
DB+MAFM4	×	0.867	0.863	0.865
DB+MAFM1_2	×	0.866	0.863	0.865
DB+MAFM1_2_3	×	**0.872**	0.860	0.866
DB+MAFM1_2_3_4	×	0.870	0.864	**0.867**
DB+MAFM1	✓	0.875	0.865	0.870
DB+MAFM2	✓	0.875	0.872	0.873
DB+MAFM3	✓	0.875	0.869	0.872
DB+MAFM4	✓	0.876	0.869	0.872
DB+MAFM1_2	✓	0.873	**0.874**	0.873
DB+MAFM1_2_3	✓	0.871	0.873	0.872
DB+MAFM1_2_3_4	✓	**0.889**	0.869	**0.879**

12) Ablation Experiments of MAFM: We add MAFM in each of the four feature mapping stages of the deep network branch. As shown in Table XIII, when SDRM is not added, the results show that the F_1 is improved when only one MAFM is added. When we arrange all MAFMs in order, F_1 increases

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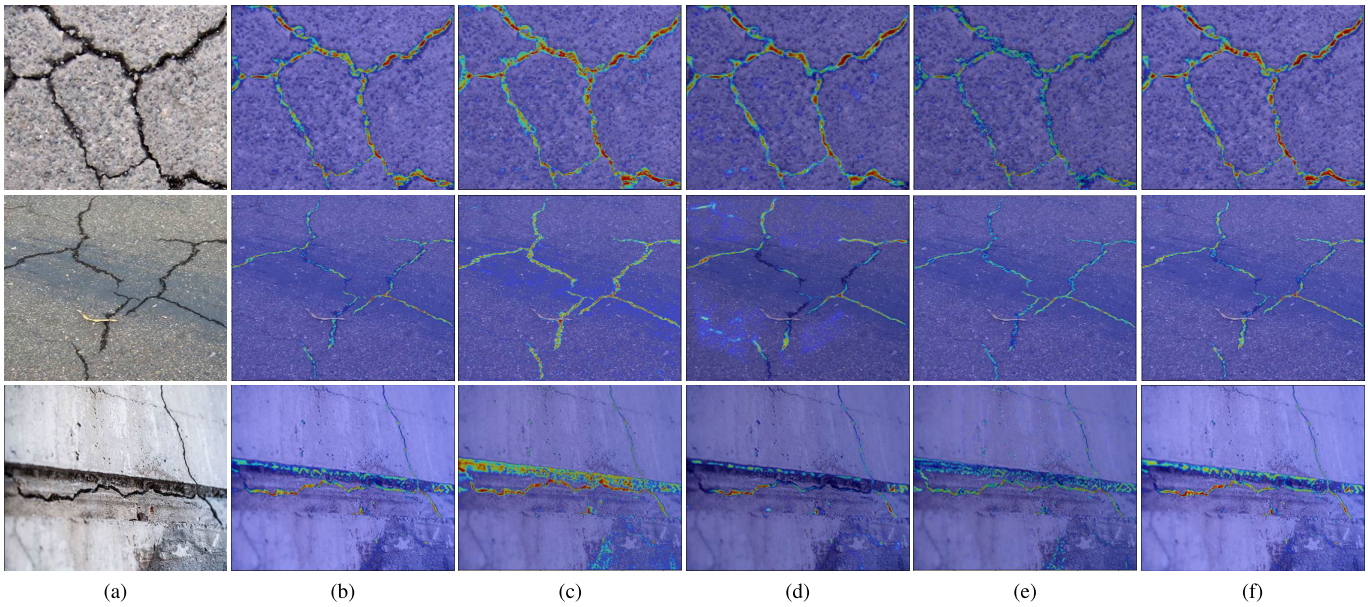

Fig. 14. Visualisation results of heat maps on proposed method. (a) Raw image. (b) The last layer of the network. (c) The spatial path of MAFM4. (d) The context path of MAFM4. (e) The last layer of shallow network. (f) The last layer of deep network.

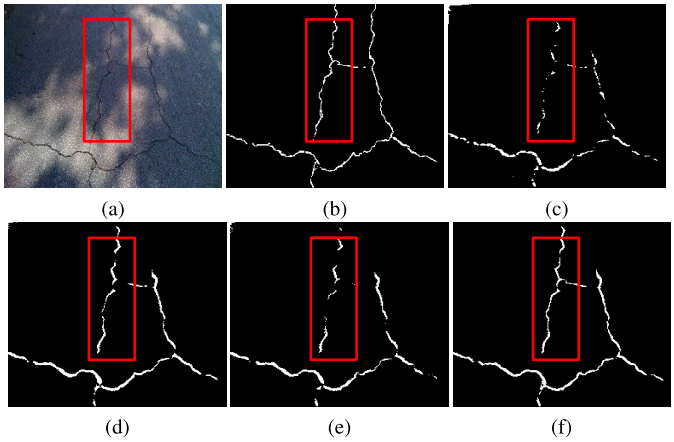

Fig. 15. The visualization of detection results. (a) Raw Image; (b) Ground Truth; (c) DB; (d) DB+SDRM; (e) DB+MAFM; (f) MFDB-Net.

TABLE XIV
THE ABLATION EXPERIMENTS OF MAFM SPATIAL PATH(S) AND CONTEXT PATH(C) ON DeepCrack

Methods	SDRM	Metrics		
		P	R	F_1
MAFM-S-C	×	0.864	0.860	0.862
MAFM+S-C	×	**0.872**	0.857	0.864
MAFM-S+C	×	0.865	**0.869**	0.867
MAFM+S+S	×	0.854	0.867	0.861
MAFM+C+C	×	0.867	0.863	0.865
MAFM+S+C	×	0.870	0.864	**0.867**
MAFM-S-C	✓	0.874	0.866	0.870
MAFM+S-C	✓	0.887	0.866	0.876
MAFM-S+C	✓	0.882	0.865	0.874
MAFM+S+S	✓	0.877	0.869	0.873
MAFM+C+C	✓	0.875	0.871	0.873
MAFM+S+C	✓	**0.889**	**0.869**	**0.879**

by 1.4% over the baseline DB. When SDRM is added, the results have a similar improvement effect to the second set of experimental results in Table XIII. SDRM combines and superimposes multiple ordinary convolutions and dilated convolutions with different dilation rates in parallel, which can increase the receptive field while keeping the size of the feature map unchanged. We conduct ablation experiments on the spatial path and semantic path that constitute MAFM. As shown in Table XIV, directly fusing the features of the bottom network and the high-level network without using the spatial path or the semantic path has the worst effect. Using only the spatial path or the semantic path to process the features of the low-level network and the high-level network will obtain better features than not doing any processing. Using the spatial path to extract the features of the low-level network and the semantic path to extract the features of the high-level network at the same time will obtain more refined features and achieve the highest F_1.

13) Ablation Experiments of the Modules: As shown in Table XV, our ablation experiments on the DeepCrack dataset are carried out to illustrate the effectiveness of our proposed method for detecting cracks. The naming rules of these four models are as follows: DB represents the proposed basic dual-branch network model, DB+SDRM represents the model containing DB and four series hole residual modules, DB+MAFM is represented by DB and four multi-attention fusion modules, DB+SDRM+MAFM represents the overall network MFDB-Net we proposed. We use DB as the baseline. From Table XV we can find that both SDRM and MAFM can achieve significant improvements in most metrics based on the DB baseline. Introducing SDRM and MAFM into DB achieves the most optimal results in our conducted experiments. This result demonstrates that the enhancements made by the two modules are effective for crack detection.

As shown in the visual heat map of Fig. 14, Fig. 14(c) and Fig. 14(d) show that the spatial path has rich spatial information, and the semantic path has precise semantic information, but less edge information. Fig. 14(e) and Fig. 14(f)

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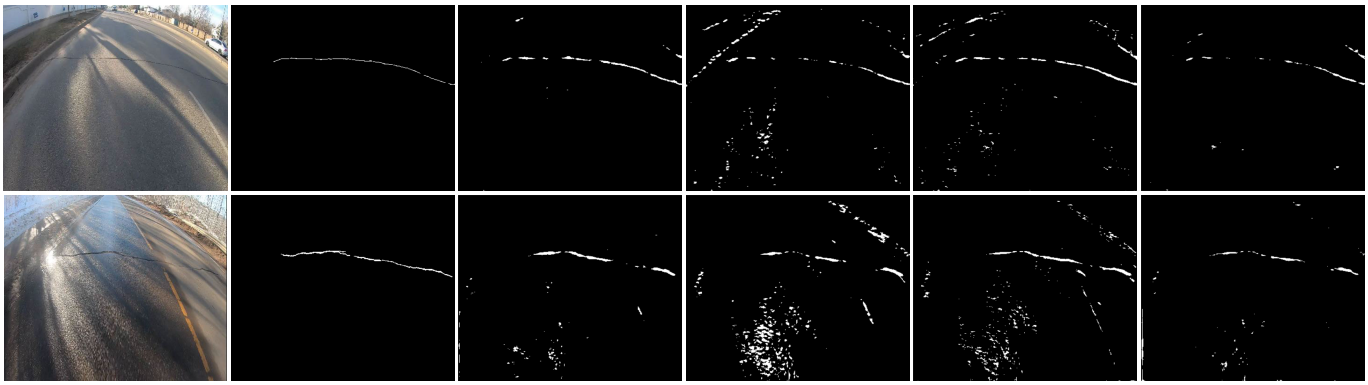

Fig. 16. The Visualisation results of MFDB on DCCE. (a) Raw image. (b) Ground truth. (c) MFDB. (d) DB. (e) DB+SDRM. (f) DB+MAFM.

TABLE XV
THE ABLATION EXPERIMENTS ON DeepCrack

Methods	Metrics					
	P	R	F_1	PA	C	$MIoU$
DB	0.827	**0.878**	0.853	0.986	**0.935**	0.864
DB+SDRM	0.867	0.865	0.866	0.988	0.923	0.876
DB+MAFM	0.870	0.864	0.867	0.988	0.929	0.876
MFDB-Net(Ours)	**0.889**	0.869	**0.879**	**0.989**	0.932	**0.886**

TABLE XVI
THE ABLATION EXPERIMENTS FOR SPECIFIC CHALLENGES ON DCCE

Methods	Metrics					
	P	R	F_1	PA	C	$MIoU$
DB	0.652	0.704	0.677	0.969	0.843	0.740
DB+SDRM	0.667	0.703	0.684	0.970	0.843	0.744
DB+MAFM	0.661	0.721	0.689	0.970	0.851	0.747
MFDB-Net(Ours)	**0.681**	**0.725**	**0.702**	**0.971**	**0.854**	**0.756**

show that the shallow network of the proposed dual-branch network contains more contour and edge information, and the deep network establishes global context connections and extracts more semantic information. These conclusions are also analyzed in Sec. III and Table IX. The proposed deep branch network is to extract deeper semantic information, which plays a dominant role in the entire network framework. The shallow branch network is to retain the spatial information related to the crack as possible, and plays an auxiliary role in the entire network framework to make up for the detailed information lost as the deep network deepens. From Fig. 15, we find that the proposed method is cumulative, processing detailed information better and better. As shown on the red box mark, MFDB can detect more continuous and fine cracks, and finally obtains an image similar to Ground Truth.

14) Ablation Experiments for Specific Challenges: To investigate the impact of specific challenges, such as shadows, occlusions, on the model performance, we perform ablation experiments on DCCE. As shown in the Fig. 16 and Table XVI, the shadow brings difficulties to crack detection, and the baseline DB has difficulty in detecting valid cracks. DB+SDRM can detect more complete cracks than DB. SDRM is applied in the decoding layer, which increases the receptive field while keeping the size of the feature map unchanged. The extracted multi-scale features are used to capture the global context information, and the crack features are more effectively retained. The anti-jamming ability of DB+MAFM against background is strengthened. MAFM is applied in the mapping of decoder layer to decoder layer, which reduces the feature mapping gap between different network layers, uses the characteristics of each network layer, and improves the ability of network to extract semantics. MFDB combines the advantages of the two proposed modules and is able to detect more complete cracks while better distinguishing non-crack pixels.

TABLE XVII
THE *Params*, AND *FPS* OF ALL METHODS ON DeepCrack

Methods	Metrics	
	$Params$	FPS
U-Net [34]	31.03M	17
DeepCrack [36]	17.72M	26
FPHBN [38]	14.97M	28
UHDN [40]	17.97M	12
CrackW-Net [42]	28.33M	36
MSCF [35]	27.51M	12
CrackFormer [37]	4.55M	9
PAF-Net [39]	12.62M	26
DSC-Net [41]	20.97M	6
CMUNeXt [43]	3.15M	31
DB(Ours)	11.53M	17
MFDB-Net(Ours)	28.69M	10

15) Model Complexity: We measure model complexity using $Params$ and FPS. It is assumed that the convolution is implemented as a sliding window and the nonlinear function consumes no computing power. For convolutional layers, $Params$ is calculated as follows,

$$Params = (C_{in}K^2 + 1)C_{out}, \tag{22}$$

where C_{in} is the number of channels of the input, C_{out} is the number of channels of the output, K is the kernel size.

The input feature is a $3\times256\times256$ tensor. From Table XVII, the proposed method has no advantage in $Params$ and FPS. Because we keep the feature resolution unchanged in the shallow network branch to avoid the loss of information caused by downsampling, which makes $Params$ larger. Compared with other methods on crack datasets, this method has better precision performance, which makes up for the shortcomings of the model in learning ability and feature extraction, and can effectively extract more complex features.

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V. Conclusion

In this paper, a dual-branch network for detecting cracks is proposed, which forms the foundation for developing two modules: a series dilated residual module and a multi-attention fusion module. The series dilated residual module is employed for extracting multiscale features and capturing global contextual details. The multi-attention fusion module optimizes crack feature information by capitalizing on the distinctive attributes of various network layers, effectively diminishing loss and redundancy. We conduct experiments on five publicly available crack datasets respectively. Experimental results indicate that our method attains optimal efficacy.

In order to enable the crack detection network model to be transplanted to mobile devices and used in real scenarios, we can study compression algorithms such as knowledge distillation and shearing in the future.

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